

Module 6.1: Lognormal Distributions and Simulation Techniques

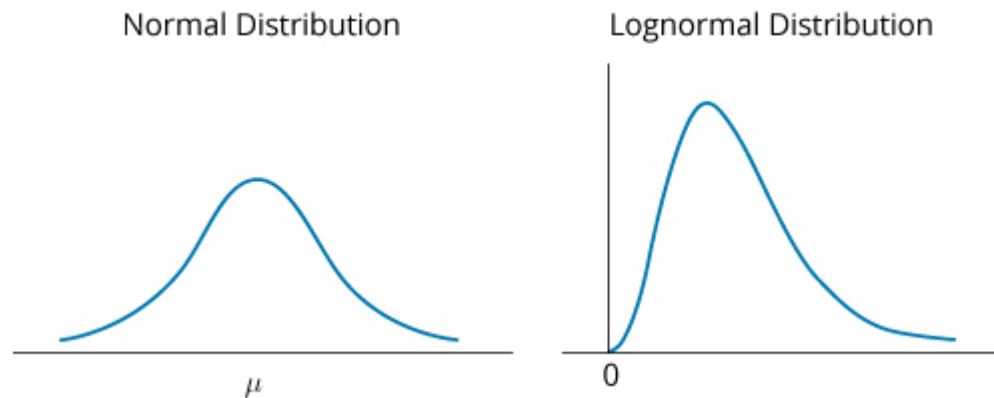
SCHWESERNOTES - BOOK 1

LOS 6.a: Explain the relationship between normal and lognormal distributions and why the lognormal distribution is used to model asset prices when using continuously compounded asset returns.

The **lognormal distribution** is generated by the function e^x , where x is normally distributed. Because the natural logarithm, $\ln$, of e^x is x , the logarithms of lognormally distributed random variables are normally distributed, thus the name.

Normal vs. Lognormal Distributions illustrates the differences between a normal distribution and a lognormal distribution.

Normal vs. Lognormal Distributions



Recall from our reading about rates and returns that we use the natural logarithm to calculate continuously compounded returns. The lognormal distribution is useful for modeling asset prices if we think of an asset's future price as the result of a continuously compounded return on its current price. That is:

$$P_T = P_0 e^{r_{0,T}}$$

where:

P_T = asset price at time T

P_0 = asset price at time 0 (today)

$r_{0,T}$ = continuously compounded return on the asset from time 0 to time T

Because continuously compounded returns are additive, we can divide the time from 0 to T into shorter periods and state that $r_{0,T}$ is the sum of the continuously compounded returns over each of these shorter periods. Then, if we assume each of these returns is normally distributed, we can state that $r_{0,T}$ is normally distributed. Even if they are not normally distributed, the central limit theorem implies that their sum ($r_{0,T}$) is approximately normally distributed. This allows us to say P_T is lognormally distributed, because it is proportional to the logarithm of a normally distributed variable.

In many of the pricing models that we will see in the CFA curriculum, we assume returns are **independently and identically distributed**. If returns are *independently distributed*, past returns are not useful for predicting future returns. If returns are *identically distributed*, their mean and variance do not change over time (a property known as *stationarity* that is important in time series modeling, a Level II topic).

LOS 6.b: Describe Monte Carlo simulation and explain how it can be used in investment applications.

Monte Carlo simulation is a technique based on the repeated generation of one or more risk factors that affect security values to generate a distribution of security values. For each of the risk factors, the analyst must specify the parameters of the probability distribution that the risk factor is assumed to follow. A computer is then used to generate random values for each risk factor based on its assumed probability distributions. Each set of randomly generated risk factors is used with a pricing model to value the security. This procedure is repeated many times (100s, 1,000s, or 10,000s), and the distribution of simulated asset values is used to draw inferences about the expected (mean) value of the security—and possibly the variance of security values about the mean as well.

As an example, consider the valuation of stock options that can only be exercised on a particular date. The main risk factor is the value of the stock itself, but interest rates could affect the valuation as well. The simulation procedure would be to do the following:

1. Specify the probability distributions of stock prices and of the relevant interest rate, as well as the parameters (e.g., mean, variance, skewness) of the distributions.
2. Randomly generate values for both stock prices and interest rates.
3. Value the options for each pair of risk factor values.
4. After many iterations, calculate the mean option value and use that as your estimate of the option's value.

Monte Carlo simulation is used to do the following:

- Value complex securities.
- Simulate the profits/losses from a trading strategy.
- Calculate estimates of value at risk (VaR) to determine the riskiness of a portfolio of assets and liabilities.
- Simulate pension fund assets and liabilities over time to examine the variability of the difference between the two.
- Value portfolios of assets that have nonnormal return distributions.

An advantage of Monte Carlo simulation is that its inputs are not limited to the range of historical data. This allows an analyst to test scenarios that have not occurred in the past. The limitations of Monte Carlo simulation are that it is fairly complex and will provide answers that are no better than the assumptions about the distributions of the risk factors and the pricing/valuation model that is used. Also, simulation is not an analytic method, but a statistical one, and cannot offer the insights provided by an analytic method.

LOS 6.c: Describe the use of bootstrap resampling in conducting a simulation based on observed data in investment applications.

Resampling is another method for generating data inputs to use in a simulation. Often, we do not (or cannot) have data for a population, and can only approximate the population by sampling from it. (For example, we may think of the observed historical returns on an investment as a sample from the population of possible return outcomes.) To conduct resampling, we start with the observed sample and repeatedly draw subsamples from it, each with the same number of observations. From these samples, we can infer parameters for the population, such as its mean and variance.

In our reading on Estimation and Inference, we will describe some of the available resampling techniques. One of these is known as **bootstrap resampling**. In bootstrap resampling, we draw repeated samples of size n from the full dataset, replacing the sampled observations each time so that they might be redrawn in another sample. We can then directly calculate the standard deviation of these sample means as our estimate of the standard error of the sample mean.

Simulation using data from bootstrap resampling follows the same procedure as Monte Carlo simulation. The difference is the source and scope of the data. For example, if a simulation uses bootstrap resampling of historical returns data, its inputs are limited by the distribution of actual outcomes.

Reading 6: Key Concepts

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LOS 6.a

If x is normally distributed, e^x follows a lognormal distribution. The lognormal distribution is useful for modeling an asset's future price as the result of a continuously compounded return on its current price.

If investment returns are independently distributed, past returns are not useful for predicting future returns.

If investment returns are identically distributed, their mean and variance do not change over time.

LOS 6.b

Monte Carlo simulation uses randomly generated values for risk factors, based on their assumed distributions, to produce a distribution of possible security values. Its limitations are that it is fairly complex and will provide answers that are no better than the assumptions used.

LOS 6.c

Bootstrap resampling involves drawing repeated samples from a sample that represents the population, replacing the sampled observations each time so that they might be redrawn in another sample. The standard deviation of these

sample means is an estimate of the standard error of the sample mean. As with all simulation techniques, its answers are no better than the assumptions used.